

4.2 Reduction of order

If y_1 and y_2 are two LI solutions of $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$ (*)

then, the general solution is

$$y = c_1 y_1 + c_2 y_2.$$

In this section, we present a method to find y_2 if a solution y_1 is known.

If y_1 is a known solution of the DE, we look for another solution y_2 in the form:

$$y_2(x) = \boxed{u(x)} y_1(x).$$

some unknown function.

Finding $u(x)$ requires solving a 1st order DE, that's why it is called reduction of order.

Suppose we divide (*) by $a_2(x)$, we get the standard form DE :

$$y'' + P(x)y' + Q(x)y = 0. \quad (\text{SF})$$

Assume that $y_1 = y_1(x)$ is a known solution of (SF).

Let's look for another solution of (SF) in the form :

$$y_2(x) = u(x) y_1(x)$$

for some (unknown) function $u(x)$.

To find $u(x)$, we substitute y_2 in the DE (SF).

$$y_2 = u y_1$$

$$\Rightarrow y_2' = u' y_1 + u y_1'$$

$$\Rightarrow y_2'' = u'' y_1 + 2u' y_1' + u y_1''$$

Substitute these in (SF) to get:

$$y_2'' + P(x) y_2' + Q(x) y_2 = 0$$

$$\left[u'' y_1 + 2u' y_1' + \underline{\underline{u y_1''}} \right] + P(x) \left[u' y_1 + \underline{\underline{u y_1'}} \right] + Q(x) \left[\underline{\underline{u y_1}} \right] = 0$$

$$\Rightarrow u \left[y_1'' + P(x) y_1' + Q(x) y_1 \right] + u'' y_1 + (2 y_1' + P(x) y_1) u' = 0$$

Zero since y_1 is
a solution

$$\Rightarrow y_1 u'' + (2 y_1' + P(x) y_1) u' = 0$$

$$\text{Let } W = u' \rightarrow W' = u''.$$

Then:

$$y_1 W' + (2y_1' + P(x)y_1)W = 0$$

This is a separable 1st order DE.

$$y_1 \frac{dW}{dx} + (2y_1' + P(x)y_1)W = 0$$

$$\Rightarrow \int \frac{dW}{W} = \int \left(-\frac{2y_1'}{y_1} - P(x) \right) dx$$

$$\Rightarrow \ln |W| = -2 \ln |y_1| - \int P(x) dx + C$$

$$\Rightarrow \ln |W| + \ln |y_1^2| = - \int P(x) dx + C$$

$$\Rightarrow \ln |W y_1^2| = - \int P(x) dx + C$$

$$\Rightarrow |w y_1^2| = e^c e^{-\int P(x) dx}$$

$$\Rightarrow w y_1^2 = \underbrace{\pm e^c}_{c_1} e^{-\int P(x) dx}$$

$$\Rightarrow w = \frac{c_1 e^{-\int P(x) dx}}{y_1^2}$$

since $w = u' \rightarrow u = \int w dx$

$$u = c_1 \int \frac{e^{-\int P(x) dx}}{y_1^2} dx + c_2$$

we need a single u so by choosing

$c_1 = 1$ and $c_2 = 0$ we get

$$u = \int \frac{e^{-\int P(x) dx}}{y_1^2} dx$$

$$\Rightarrow y_2 = u y_1$$

$$\Rightarrow y_2 = y_1 \int \frac{e^{-\int P(x) dx}}{y_1^2} dx$$

Let's verify that the functions y_1 and y_2 are L.I.

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} y_1 & u y_1 \\ y_1' & u' y_1 + u y_1' \end{vmatrix} \quad (\text{Wronskian})$$

$$W = u' y_1^2 + u y_1 y_1' - u y_1 y_1'$$

$$W = u' y_1^2 \quad \text{since } u' = \frac{e^{-\int P(x) dx}}{y_1^2}$$

$$\text{then: } W = e^{-\int P(x) dx} \neq 0 \quad \text{for all } x$$

$\Rightarrow$ they are L.I.

Conclusion

If y_1 is a known solution
of

$$\underbrace{y'' + P(x)y' + Q(x)y = 0}_{\text{Standard form DE}}$$

then :

$$y_2 = y_1 \int \frac{e^{-\int P(x)dx}}{y_1^2} dx$$

is also a solution of the DE.

The functions y_1 and y_2 are
linearly independent.

The general solution is $y = c_1 y_1 + c_2 y_2$.

Note : We can always scale y_2 .

For example, if we get that $y_2 = 5x^2$.

We can scale it and use $y_2 = x^2$.

We can do that since the general

solution is $y = c_1 y_1 + c_2 y_2$.

The 5 can be absorbed in the c_2 .

$$y = c_1 y_1 + c_2 (5x^2)$$

and

$$y = c_1 y_1 + c_2 x^2$$

are equivalent.

ex: Consider the DE

$$x^2 y'' + 2xy' - 6y = 0.$$

- (a) verify that $y_1 = x^2$ is a solution.
- (b) solve the IVP with initial conditions
 $y(1) = 4$ and $y'(3)$.
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(a) $y_1 = x^2 \rightarrow y_1' = 2x \rightarrow y_1'' = 2.$

Plug these in the DE to get:

$$x^2 y_1'' + 2x y_1' - 6 y_1$$

$$= x^2 [2] + 2x [2x] - 6 [x^2]$$

$$= 2x^2 + 4x^2 - 6x^2 = 0 \quad \checkmark$$

$\therefore y_1$ is a solution.

(b) The standard form DE is :

$$y'' + \left(\frac{2}{x} \right) y' - \frac{6}{x^2} y = 0$$

$\downarrow$
 $P(x)$

$$y_2 = y_1 \int \frac{e^{-\int P(x) dx}}{y_1^2} dx$$

$$y_2 = x^2 \int \frac{e^{-\int \frac{2}{x} dx}}{(x^2)^2} dx$$

$$y_2 = x^2 \int \frac{e^{-2 \ln x}}{x^4} dx$$

$$y_2 = x^2 \int \frac{e^{\ln(x^{-2})}}{x^4} dx$$

$$y_2 = x^2 \int \frac{x^{-2}}{x^4} dx$$

$$y_2 = x^2 \int x^{-6} dx$$

$$y_2 = x^2 \left(\frac{x^{-5}}{-5} \right)$$

$$y_2 = -\frac{1}{5} x^{-3}$$

Let's scale and use :

$$y_2 = x^{-3}.$$

We have two LI solutions :

$$y_1 = x^2 \text{ and } y_2 = x^{-3}.$$

The general solution is $y = c_1 y_1 + c_2 y_2$.

$$\Rightarrow y = c_1 x^2 + c_2 x^{-3}.$$

Let's use the initial conditions to
Find c_1 and c_2 .

$$\text{We need } y' = 2c_1x - 3c_2x^{-4}.$$

$$y(1) = 4 \longrightarrow c_1 + c_2 = 4$$

$$y'(1) = 3 \longrightarrow 2c_1 - 3c_2 = 3$$

Solve the linear system to get:

$$c_1 = 3 \text{ and } c_2 = 1.$$

The unique solution of the IVP is

$$y = 3x^2 + x^{-3}$$

$$\text{or } y = 3x^2 + \frac{1}{x^3}.$$